OO-C-851a
May 11, 1967
SUPERSEDING
Fed. Spec. OO-C-851
April 4, 1956

FEDERAL SPECIFICATION

CUTTER BIT, TOOL HOLDER (SOLID HIGH-SPEED STEEL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cutter bit blanks and straight cut-off blades of high-speed tool steel.

1.2 Classification.

1.2.1 Types and styles. Cutter bits covered by this specification shall be of the following types and styles, as specified (see 6.2):

Type I—Cutter bit blanks.
- Style 1—Square.
- Style 2—Rectangular.

Type II—Straight cut-off blades.
- Style 1—Straight side clearance.
- Style 2—For engine lathe tool holders.

.2.2 Grades and classes. Cutter bits shall be of the following grades and classes, as specified (see 6.2):

Grade A—General purpose.
- Class T-1.
- Class T-2.
- Class T-7.
- Class M-1.
- Class M-2.
- Class M-3.
- Class M-10.

Grade B—Heavy duty.
- Class T-4.
- Class T-5.
- Class T-6.
- Class T-8.
- Class T-15.
- Class M-30.
- Class M-34.
- Class M-41.
- Class M-42.
- Class M-43.
- Class M-44.

1.2.3 Finishes. Cutter bits shall have the following finishes, as specified (see 6.2).

Finish a—Ground on all sides.

Finish b—Sandblasted on all sides.

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2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Cutter bits shall be high-speed tool steel alloy.

3.2 Chemical composition. The chemical composition shall conform to table I for the class of steel specified (see 6.2), when tested as specified in 4.4.2. When grades only are specified, the chemical composition of any of the classes shown under 1.2.2 may be furnished, as applicable.

TABLE I. Chemical composition

Class	Elements							
	Carbon		Manganese	Silicon	Phosphorus	Sulfur	Chromium	
	(Min.)	(Max.)	(Max.)	(Max.)	(Max.)	(Max.)	(Min.)	(Max.)
	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
T-1	0.70	0.76	0.40	0.40	0.03	0.03	3.75	4.25
T-2	.80	.86	.40	.40	.03	.03	3.75	4.50
T-4	.70	.76	.40	.40	.03	.03	3.75	4.50
T-5	.75	.81	.40	.40	.03	.03	4.00	4.75
T-6	.75	.81	.40	.40	.03	.03	4.00	4.75
T-7	.70	.76	.40	.40	.03	.03	3.75	4.25
T-8	.76	.83	.40	.40	.03	.03	3.75	4.50
T-15	1.45	1.60	.40	.40	.03	.03	4.00	5.00
M-1	0.77	0.83	.40	.40	.03	.03	3.50	4.25
M-2	.79	.86	.40	.40	.03	.03	3.90	4.40
M-3	1.00	1.25	.40	.40	.03	.03	3.75	4.40
M-10	0.84	0.90	.40	.40	.03	.03	3.75	4.50
M-30	.79	.85	.40	.40	.03	.03	3.50	4.25
M-34	.84	.90	.40	.40	.03	.03	3.50	4.25
M-41	1.00	1.15	.40	.40	.03	.03	4.00	4.75
M-42	1.00	1.15	---	---	---	---	3.25	4.00
M-43	1.15	1.25	---	---	---	---	3.25	3.75
M-44	1.00	1.15	---	---	---	---	3.75	4.25

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TABLE I. Chemical composition (cont'd.)

Class	Elements										
	Tungsten		Vanadium		Molybdenum		Cobalt		Copper	Nickel	Tin
	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	(Max.)	(Max.)	(Max.)
	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
T-1	17.50	18.75	0.90	1.30	---	---	---	---	0.25	0.25	0.05
T-2	17.50	18.75	1.90	2.30	0.60	1.00	---	---	.25	.25	.05
T-4	17.50	18.75	1.00	1.40	.50	1.00	4.50	5.50	.25	.25	.05
T-5	17.75	19.00	1.80	2.20	.50	1.00	7.50	9.50	.25	.25	.05
T-6	19.00	22.00	1.60	2.00	.50	1.00	11.00	14.00	.25	.25	.05
T-7	13.50	14.50	1.80	2.20	---	---	---	---	.25	.25	.05
T-8	13.50	14.50	1.80	2.20	.50	1.00	4.50	5.50	.25	.25	.05
T-15	12.00	12.75	4.75	5.50	---	1.00	4.75	5.50	.25	.25	.05
M-1	1.30	2.00	1.00	1.30	8.00	9.20	---	---	.25	.25	.05
M-2	6.00	6.75	1.70	2.10	4.75	5.25	---	---	.25	.25	.05
M-3	5.50	6.50	2.30	3.30	5.25	6.00	---	---	.25	.25	.05
M-10	---	---	1.80	2.20	7.75	9.00	---	---	.25	.25	.05
M-30	1.30	2.30	1.00	1.40	7.75	9.00	4.50	5.50	.25	.25	.05
M-34	1.20	2.30	1.90	2.30	8.00	9.20	7.75	8.75	.25	.25	.05
M-41	6.50	7.75	1.60	2.20	3.25	4.25	4.75	5.25	---	---	---
M-42	1.25	1.80	1.00	1.30	8.75	9.75	7.25	8.75	---	---	---
M-43	1.50	1.75	1.75	2.00	8.00	8.75	7.50	8.25	---	---	---
M-44	4.50	5.25	2.00	2.25	5.50	6.25	11.00	12.00	---	---	---

3.3 Heat treatment. The bits shall be hardened and multiple tempered to a Rockwell C hardness of not less than 63 when tested as specified in 4.4.1.

3.3.1 Texture and microstructure. The texture and microstructure of the cutter bits, after heat treatment, shall conform to the requirements specified in 4.4.2.

3.4 Finish a (ground on all sides). The cutter bits shall be ground on all sides. Grinding shall be sufficient to remove all decarburization.

3.4.1 Tolerance for ground cutter bits. A plus or minus tolerance of 0.002 inch will be permitted on cross-sectional dimensions, and a minus tolerance of 1/16 inch will be permitted on the length of cutter bits when measured as shown by "L" on figures 1, 2, 3, and 4.

3.5 Finish b (sandblasted on all sides). Cross-sectional dimension shall be within the tolerance specified in table II. Tolerance on lengths of sandblasted cutter bits shall be as specified in 3.4.1 for ground cutter bits. The bits shall not be decarburized beyond the limits shown in table II.

TABLE II. Dimensional tolerances for sandblasting bits

Dimensions	Permissible variation (plus or minus)	Depth of decarburization (maximum)
Inches	Inch	Inch
1/16 to 1/2	0.010	0.015
1/2 to 1	.015	.025
1 to 1 1/2	.031	.035
1 1/2 to 2	.046	.047

3.6 Identification marking. The bits shall be marked with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.7 Type I, cutter bit blanks.

3.7.1 Style 1, square. Style 1 bits shall have a square cross section with the ends beveled parallel to each other at an angle of 20 ± 10°. They shall be similar to figure 1 and shall be furnished in sizes conforming to table III as specified (see 6.2).

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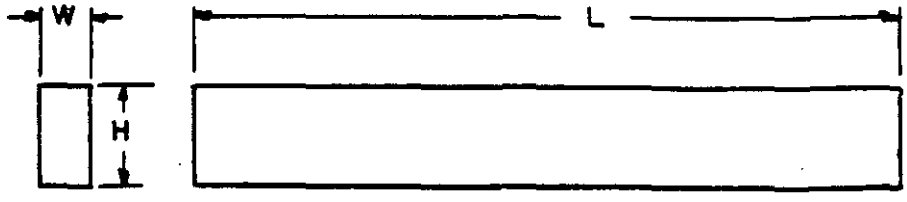
SH 8558

FIGURE 1. Type I, style 1, square cutter bit blank.

TABLE III. Dimensions of square cutter bit blanks

Cross-section square	Length L
Inches	Inches
3/16	2½
[illegible]	2½
	2½
	3
[illegible]	3½
½	4
⅝	4½
¾	5
1	7
1¼	7
1½	10

3.7.2 Style 2, rectangular. Style 2 bits shall have a rectangular cross section with the ends squared, shall be similar to figure 2, and shall be furnished in sizes conforming to table IV, as specified (see 6.2).



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FIGURE 2. Type I, style 2, rectangular cutter bit blank.

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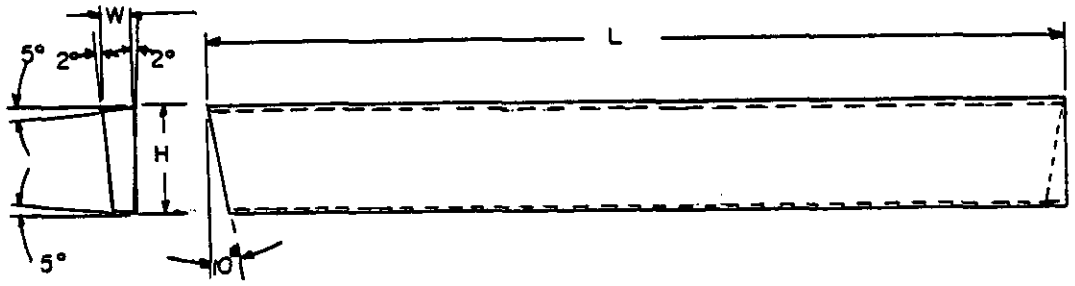
TABLE IV. Dimensions of rectangular cutter bit blanks

Cross section		Length
Width W	Height H	L
Inch	Inches	Inches
1/4	1/2	4
5/16	5/8	4 1/2
3/8	3/4	5
1/2	3/4	5 1/4
1/2	1	5 1/2
1/2	1	7
5/8	1 1/4	6 1/2
5/8	1 1/4	8
3/4	1 1/2	6 1/2
3/4	1 1/2	9
1	1 1/2	10
1	2	12

3.8 Type II, straight cut-off blades.

3.8.1 Style 1, straight side clearance. Style 1 bits shall have a cross section in the shape of a trapezoid. Both narrow edges of the blade shall be beveled at a 5° angle converging toward a common point, to permit clamping in the holder. The sides of the blade shall be tapered 1° to 2° on each side to provide clearance. One end of the blade shall be tapered approximately 10° for end clearance angle and the other may be square or tapered for end clearance, but in either case the length shall be considered as the length on the top of the blade as shown on figure 3 and shall be furnished in sizes conforming to table V (see 6.2).

3.8.2 Style 2, for engine lathe tool holders. Style 2 bits shall have a cross section as shown on figure 4. Both narrow edges of the blade shall be beveled at a 15° ± 1° angle converging toward a common point



SH 8560

FIGURE 3. Type II, style 1, straight cut-off blade with straight side clearance.

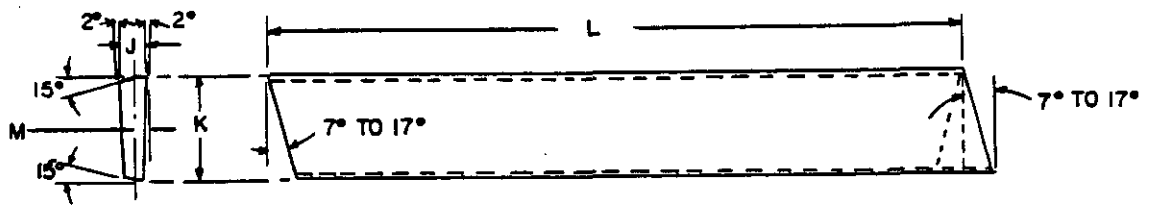
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TABLE V. Dimensions of straight cut-off blades with straight side clearance

Width W	Height H	Length L
Inch	Inches	Inches
1/16	13/32	3
1/16 [1]	1/2	4 1/2
1/16	11/16	5
1/16	13/16	6
5/64	1/2	4 1/2
3/32 [1]	1/2	4 1/2
3/32	11/16	5
3/32	13/16	6
3/32	1	6 1/2
1/8 [1]	1/2	4 1/2
1/8 [1]	11/16	5
1/8	13/16	6
1/8	1	6 1/2
5/32 [1]	11/16	5
5/32	13/16	6
3/16 [1]	11/16	5
3/16	13/16	6
3/16 [1]	7/8	6
3/16	1	6 1/2
3/16 [1]	1 1/8	6 1/2
1/4	13/16	6
1/4 [1]	7/8	6
1/4	1	6 1/2
1/4 [1]	1 1/8	6 1/2

[1] Preferred blade sizes for future design and development.

and the top shall be flatted for the width shown as "M" on figure 4 and table VI. The sides of the blade shall be tapered 2° ± 3/4° on each side to provide clearance. One end of the blade shall be tapered 7° to 17° for end clearance. The other end may be square or tapered in either direction as shown on figure 4, but in any case the length shall be considered as the length on top of the blade as shown on figure 4. Blades ordered under the nominal sizes shown in table VI shall conform to the dimensions for their respective sizes.



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FIGURE 4. Type II, style 2, cut-off blades for engine lathe tool holders.

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TABLE VI. Dimensions of cut-off blades for engine lathe tool holders

Thickness of blade	Height of blade	Length of blade	Flat width on blade	
J	K	L	M	
Nom.	Nom.	Nom.	Min.	Max.
Inch	Inches	Inches	Inch	Inch
3/32	1/2	4 1/2	0.046	0.063
3/32	5/8	5	.046	.063
1/8	3/4	5	.062	.081
1/8	7/8	6	.062	.081
3/16	1	6 1/2	.093	.114
3/16	1 3/8	7	.093	.114
1/4	1 1/4	7	.141	.154

3.9 Workmanship. Workmanship of the bits shall be of the quality prevailing among manufacturers normally producing bits of the types, grades, classes, and styles listed herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Inspection lot. All cutter bits of the same type, style, grade, class, and finish presented at one time shall be considered a lot for the purpose of inspection.

4.2.2 Sampling for visual and dimensional examination. A random sample of bits shall be selected from each inspection lot in accordance with MIL-STD-105, inspection levels II. The acceptable quality level (AQL) shall be equal to 1.5 percent defective.

4.2.3 Sampling for hardness tests. A random sample of bits shall be taken from each inspection lot in accordance with table VII and shall be subjected to the hardness tests specified in 4.4.1.

TABLE VII. Sampling for hardness test AQL (approx.) = 1.0 percent defective

Number of bits in inspection lot	Number of bits in sample	Acceptance number (defectives)	Rejection number (defectives)
15 or under	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	0	1
301 to 500	25	0	1
501 to 800	35	1	2
801 and over	50	1	2

4.2.4 Sampling for texture and microstructure test, and chemical analysis. When specified (see 6.2), a random sample of bits shall be taken from each inspection lot in accordance with table VIII for the texture and microstructure test, and the chemical analysis specified in 4.4.2.

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TABLE VIII. Sampling for texture, microstructure test and chemical analysis

Number of bits in inspection lot	Number of bits in sample [1]
110 or less	2
111 to 500	3
501 and over	4

[1] The number of samples shall be increased where the weight is less than 6 ounces to a number weighing at least 6 ounces.

4.3 Visual and dimensional examination. Each of the sample bits taken in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Any bits in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective bits in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 Hardness test. Each of the sample bits taken in accordance with table VII shall be tested on a Rockwell hardness testing machine with a "Brale" point and a 150 kilogram load to determine conformance with 3.3. If the number of nonconforming bits in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4.2 Tests for texture, microstructure, and chemical analysis. One or more of the sample bits taken in accordance with table VIII shall be nicked and fractured. The fractured surface shall show a uniform texture, fine grain, and no harmful defects shall be present such as excessive streaks, segregations, laminations, or fish scale. Specimens shall be cut from the fractured bits, polished, etched in 2 percent nital solution (2 percent nitric acid-ethanol) and examined at a magnification of 1000 diameters. The microstructure shall show a completely tempered martensitic matrix, a fine grain, and evidence of prior austenitizing and quenching conditions which indicate a satisfactory solution of alloy carbides with freedom from any indication of the formation of harmful liquid phase. The sample bits including the fractured samples, if required, shall be annealed and about 2 ounces of millings or turnings secured for chemical analysis to determine conformance with 3.2. If any sample fails any test, this shall be cause for rejection of the lot represented by the sample.

4.5 Examination of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the document referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.2), the tool holder cutter bits shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing shall be A, B, or C, as specified (see 6.2).

5.2 Marking.

5.2.1 Military agencies. Marking shall be in accordance with MIL-STD-129.

5.2.2 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The cutter bits and tool holders described herein are intended to be used in tool posts, boring bars, etc.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style, grade, class (when applicable see 3.2), finish, and size of bits required (see 1.2, applicable table of dimensions).
(c) Texture and microstructure test and chemical analysis, if required (see 4.2.4).
(d) Levels of preservation, packaging, and packing (see 5.1).

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6.3 The high-speed steels are basically divided into two classes by the major alloying elements entering into their composition, that is, tungsten and molybdenum. All high-speed steels in which tungsten is the major alloying element are designated by the letter "T" as the first character in the symbol; all high-speed steels in which molybdenum is the major alloying element are designated by the letter "M" as the first character in the symbol.

Classes of steel	Recommended use
T-1, T-2, T-7 M-1, M-2, M-3, M-10	General purpose.
T-4, T-5, T-6, T-8, T-15, M-30, M-34, M-41, M-42, M-43, M-44	Heavy cuts and hard material.

6.4 The technical requirements of this specification are essentially the same as those contained in the American Standard Association Pamphlets No. B5, 22-1950 "Single Point Tools and Tool Posts" and B5. 21-1949 "Staight Cut-Off Blades", issued by the United States of America Standard Institute, 10 East 40th St., New York, N. Y., 10016.

6.5 For Navy, Naval Ship Systems Command purchases, all cutter bits shall be finish "a".

MILITARY INTEREST:

Custodians:
Army—WC
Navy—SH
Air Force—84

Review activities:
Army—WC, MO
Navy—SH
Air Force—84

User activities:
Navy—AS, YD, MC

Preparing activity:
Navy—SH

CIVIL AGENCY INTEREST:
GSA-FSS

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